

WEEK - 5

HEALTHCARE DATA CLEANING & DATA MODELING

Introduction :

In Week 5, the healthcare dataset was loaded into Power BI and prepared for analysis. The data was cleaned and pre-processed using Power Query. Three separate tables, namely Dim_Patient, Dim_Admission, and Billing_table, were created. Patient ID and Admission ID were generated and used to connect the tables. Finally, relationships were created between the tables in the Model view.

Objectives :

The main objectives of Week 5 were:

- To load the healthcare dataset into Power BI.
- To perform data cleaning and pre-processing.
- To create an Age Group column.
- To create separate dimension and billing tables.
- To create Patient ID and Admission ID.
- To merge the required tables using the created IDs.
- To create relationships between the tables.

Data Pre-Processing :

The dataset was first opened in Power Query Editor for data pre-processing. The data was checked and cleaned before creating the required tables.

The following pre-processing operations were performed:

- Unwanted spaces were removed using Trim.

- Unwanted non-printable characters were removed using Clean.
- Text values were formatted using Capitalize Each Word.
- Appropriate data types were assigned to the columns.
- The required columns were selected for creating the individual tables.

	Name	Age	Gender	Blood Type	Medical Condition	Date of Adm
1	Bobby Jackson	30	Male	B-	Cancer	
2	Leslie Terry	62	Male	A+	Obesity	
3	Danny Smith	76	Female	A-	Obesity	
4	Andrew Watts	28	Female	O+	Diabetes	
5	Adrienne Bell	43	Female	AB+	Cancer	
6	Emily Johnson	36	Male	A+	Asthma	
7	Edward Edwards	21	Female	AB-	Diabetes	
8	Christina Martinez	20	Female	A+	Cancer	
9	Jasmine Aguilar	82	Male	AB+	Asthma	
10	Christopher Berg	58	Female	AB-	Cancer	
11	Michelle Daniels	72	Male	O+	Cancer	
12	Aaron Martinez	38	Female	A-	Hypertension	
13	Connor Hansen	75	Female	A+	Diabetes	
14	Robert Bauer	68	Female	AB+	Asthma	
15	Brooke Brady	44	Female	AB+	Cancer	
16	Ms. Natalie Gamble	46	Female	AB-	Obesity	
17	Haley Perkins	63	Female	A+	Arthritis	
18	Mrs. Jamie Campbell	38	Male	AB-	Obesity	
19	Luke Burgess	34	Female	A-	Hypertension	
20	Daniel Schmidt	63	Male	B+	Asthma	

Creation of Age Group Column :

An Age Group column was created using a Conditional Column. The age values were grouped into different age categories based on the given conditions.

This column was created to categorize patients according to their age and make the patient data easier to analyze.

Column Name	Operator	Value	Output
Age	is less than or equal to	12	Child
Age	is less than or equal to	19	Teenager
Age	is less than or equal to	39	Younger Adult
Age	is less than or equal to	59	Elder Adult

Else Clause: Senior Adult

Creation of Dim_Patient Table :

A separate Dim_Patient table was created for storing patient-related information.

The table contains the following columns:

- Patient_ID
- Name
- Age
- Gender
- Blood Type
- Age Group

Duplicate patient records were removed before creating the ID. An Index Column starting from 1 was then added and renamed as Patient_ID.

	Patient_ID	Name	Age	Gender	Blood Type	Age Group
1		Bobby Jackson	30	Male	B-	Younger Adu
2		Leslie Terry	62	Male	A+	Senior Adult
3		Danny Smith	76	Female	A-	Senior Adult
4		Andrew Watts	28	Female	O+	Younger Adu
5		Adrienne Bell	43	Female	AB+	Elder Adult
6		Emily Johnson	36	Male	A+	Younger Adu
7		Edward Edwards	21	Female	AB-	Younger Adu
8		Christina Martinez	20	Female	A+	Younger Adu
9		Jasmine Aguilar	82	Male	AB+	Senior Adult
10		Christopher Berg	58	Female	AB-	Elder Adult
11		Michelle Daniels	72	Male	O+	Senior Adult
12		Aaron Martinez	38	Female	A-	Younger Adu
13		Connor Hansen	75	Female	A+	Senior Adult
14		Robert Bauer	68	Female	AB+	Senior Adult
15		Brooke Brady	44	Female	AB+	Elder Adult
16		Ms. Natalie Gamble	46	Female	AB-	Elder Adult
17		Haley Perkins	63	Female	A+	Senior Adult
18		Mrs. Jamie Campbell	38	Male	AB-	Younger Adu
19		Luke Burgess	34	Female	A-	Younger Adu
20		Daniel Schmidt	63	Male	B+	Senior Adult

Creation of Dim_Admission Table :

A separate Dim_Admission table was created for storing admission-related information.

The table contains:

- Admission_ID

- Medical Condition
- Doctor
- Hospital
- Room Number
- Admission Type
- Medication
- Test Results

Duplicate records were removed before creating the ID. An Index Column starting from 1 was added and renamed as Admission_ID.

The screenshot shows a data table with the following columns: Admission_ID, Medical Condition, Doctor, Hospital, Room Number, and Admission Type. The table contains 20 rows of data. The 'Admission_ID' column is an index starting from 1. The 'Medical Condition' column lists various ailments like Cancer, Obesity, Diabetes, etc. The 'Doctor' column lists names like Matthew Smith, Samantha Davies, etc. The 'Hospital' column lists names like Sons And Miller, Kim Inc, etc. The 'Room Number' column lists numbers like 328, 265, etc. The 'Admission Type' column lists types like Urgent, Emergent, Elective, etc.

Admission_ID	Medical Condition	Doctor	Hospital	Room Number	Admission Type
1	Cancer	Matthew Smith	Sons And Miller	328	Urgent
2	Obesity	Samantha Davies	Kim Inc	265	Emergent
3	Obesity	Tiffany Mitchell	Cook Plc	205	Emergent
4	Diabetes	Kevin Wells	Hernandez Rogers And Vang	450	Elective
5	Cancer	Kathleen Hanna	White-White	458	Urgent
6	Asthma	Taylor Newton	Nunez-Humphrey	389	Urgent
7	Diabetes	Kelly Olson	Group Middleton	389	Emergent
8	Cancer	Suzanne Thomas	Powell Robinson And Valdez	277	Emergent
9	Asthma	Daniel Ferguson	Sons Rich And	316	Elective
10	Cancer	Heather Day	Padilla-Walker	249	Elective
11	Cancer	John Duncan	Schaefer-Porter	394	Urgent
12	Hypertension	Douglas Mayo	Lyons-Blair	288	Urgent
13	Diabetes	Kenneth Fletcher	Powers Miller, And Flores	134	Emergent
14	Asthma	Theresa Freeman	Rivera-Gutierrez	309	Urgent
15	Cancer	Roberta Stewart	Morris-Arellano	182	Urgent
16	Obesity	Maria Dougherty	Cline-Williams	465	Elective
17	Arthritis	Erica Spencer	Cervantes-Wells	114	Elective
18	Obesity	Justin Kim	Torres, And Harrison Jones	449	Urgent
19	Hypertension	Justin Moore Jr.	Houston Plc	260	Elective
20	Asthma	Denise Galloway	Hammond Ltd	465	Elective

Creation of Billing_table :

A separate Billing_table was created to store billing and admission-related information.

The table contains:

- Patient_ID
- Admission_ID
- Date of Admission
- Discharge Date
- Billing Amount

- Insurance Provider

The required IDs were brought into the Billing table using the Merge Queries option.

Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Patient_ID	Admission_ID
20 August 2019	Medicare	₹ 33,643.3273	26 August 2019	2	2
18 November 2020	Medicare	₹ 37,909.7824	18 December 2020	4	4
03 November 2020	Medicare	₹ 19,580.8723	15 November 2020	7	7
19 April 2020	Medicare	₹ 12,576.7956	22 April 2020	11	11
13 August 2023	Medicare	₹ 7,999.5869	05 September 2023	12	12
15 May 2020	Medicare	₹ 33,211.2954	08 June 2020	25	25
15 May 2020	Medicare	₹ 19,746.832	20 May 2020	26	26
18 May 2023	Medicare	₹ 17,993.2262	15 June 2023	33	33
24 May 2023	Medicare	₹ 21,185.9535	03 June 2023	34	34
07 January 2021	Medicare	₹ 45,453.4185	26 January 2021	36	36
17 January 2021	Medicare	₹ 32,973.9408	29 January 2021	40	40
23 August 2020	Medicare	₹ 42,684.5589	08 September 2020	45	45
12 October 2022	Medicare	₹ 45,585.8866	02 November 2022	46	46
05 April 2024	Medicare	₹ 30,590.5418	22 April 2024	51	51
08 May 2020	Medicare	₹ 19,183.1689	01 June 2020	65	65
09 December 2019	Medicare	₹ 8,320.6608	21 December 2019	66	66
21 November 2019	Medicare	₹ 10,107.1877	08 December 2019	67	67
30 January 2020	Medicare	₹ 20,227.8637	12 February 2020	70	70
12 May 2023	Medicare	₹ 10,283.7169	10 June 2023	85	85
17 June 2022	Medicare	₹ 25,302.1126	28 June 2022	91	91
19 August 2019	Medicare	₹ 25,503.6738	26 August 2019	96	96
09 July 2019	Medicare	₹ 31,516.1007	05 August 2019	104	104
15 January 2024	Medicare	₹ 40,782.9278	07 February 2024	106	106

Merging Tables and Creating IDs :

The dimension tables were connected with the Billing table using Merge Queries.

First, the required patient details were matched with the Dim_Patient table and the Patient_ID was expanded into the Billing table.

Similarly, the admission-related details were matched with the Dim_Admission table and the Admission_ID was expanded into the Billing table.

This allowed the Billing table to contain the correct Patient_ID and Admission_ID values.

Queries [4]

healthcare_dataset

Dim_Patient

Dim_Admission

Billing_Table

Table.SelectColumns(#"Reordered Columns",{"Patient_ID", "Admission_ID", "Date of Admission", "Insurance Provider", "Billing Amount", "Discharge Date"})

Query Settings

PROPERTIES

Name

Billing_Table

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Conditional Column

Capitalized Each Word

Cleaned Text

Trimmed Text

Changed Type1

Merged Queries

Expanded Dim_Patient

Merged Queries1

Expanded Dim_Admission

Reordered Columns

Removed Other Columns

Patient_ID	Admission_ID	Date of Admission	Insurance Provider	Billing Amount	Discharge Date
1	1	31-01-2024	Blue Cross	18,856.28	
2	2	20-09-2019	Medicare	33,643.33	
3	3	22-09-2022	Aetna	27,955.10	
4	4	18-11-2020	Medicare	37,909.78	
5	5	19-09-2022	Aetna	14,238.32	
6	6	20-12-2023	Unitedhealthcare	48,145.11	
7	7	03-11-2020	Medicare	19,580.87	
8	8	28-12-2021	Cigna	45,820.46	
9	9	01-07-2020	Cigna	50,119.22	
10	10	23-05-2021	Unitedhealthcare	19,784.63	
11	11	19-04-2020	Medicare	12,576.80	
12	12	13-08-2023	Medicare	7,999.59	
13	13	12-12-2019	Cigna	43,282.28	
14	14	22-05-2020	Unitedhealthcare	33,207.71	
15	15	08-10-2021	Unitedhealthcare	40,701.60	
16	16	01-10-2023	Blue Cross	12,263.36	
17	17	23-06-2020	Unitedhealthcare	24,499.85	
18	18	08-03-2020	Cigna	17,440.47	
19	19	04-03-2021	Blue Cross	18,843.02	
20	20	15-11-2022	Cigna	23,762.20	

Data Model and Relationships :

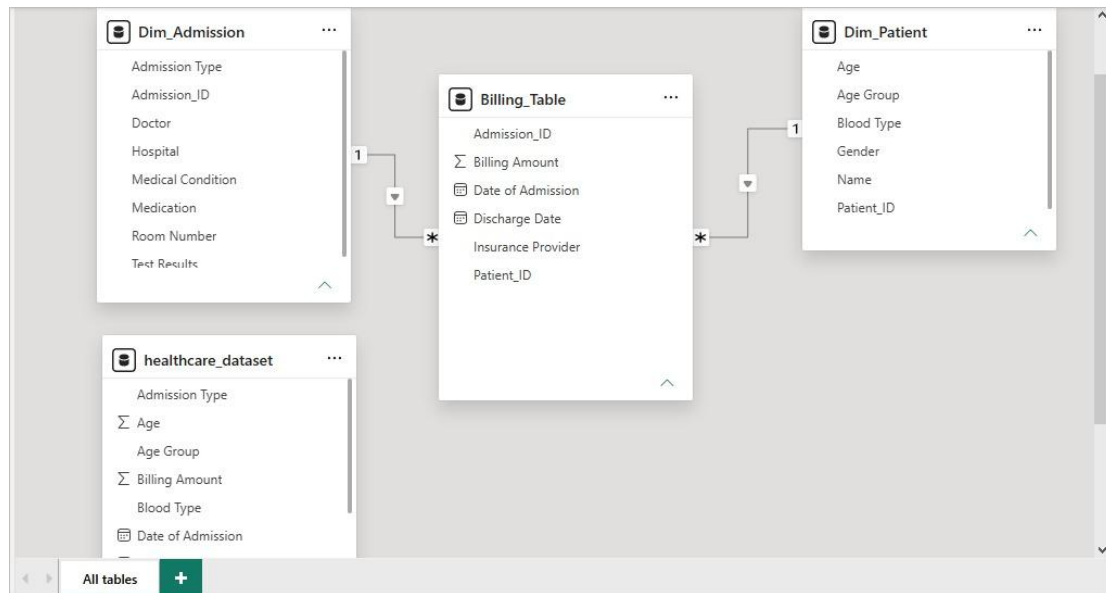
After completing the table creation and merging process, the data was loaded into the Power BI Model view.

The following relationships were created:

- Billing_table[Patient_ID] to Dim_Patient[Patient_ID]
- Billing_table[Admission_ID] to Dim_Admission[Admission_ID]

Both relationships use Many-to-One cardinality. The Billing table contains multiple records, while the dimension tables contain unique IDs. The cross filter direction is set to Single for both relationships.

The final data model connects the Billing table with the Dim_Patient and Dim_Admission tables.



Conclusion :

In Week 5, the healthcare dataset was successfully loaded and pre-processed in Power BI. Data cleaning operations such as Trim, Clean and Capitalize Each Word were performed. An Age Group column was created, and three tables **Dim_Patient**, **Dim_Admission** and **Billing_table** were created. **Patient_ID** and **Admission_ID** were generated and merged into the **Billing** table. Finally, the required one-to-many relationships were created between the tables. This completed the data preparation and data modelling process for the Week 5 task.